

Paul Carlson

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Education

Rensselaer Polytechnic Institute

Bachelor of Computer Science & Bachelor of Information Technology and Web Science

GPA: 3.60

Relevant Coursework: Intro to Algorithms, Data Structures, Principles of Software, Web Science Systems Development, Foundations of Computer Science, Computer Architecture & Operating Systems

Aug. 2024 – May 2028

Troy, NY

Experience

East-SouthEast, LLC

Web & Geospatial Technology Intern

May 2025 – Dec 2025

Chatham, MA

- Architected and built a professional-grade web mapping application from a legacy system, reducing update deployment time across nine towns from hours to minutes.
- Collected precise property and topographic survey data in the field using GPS receivers and total stations.
- Assisted with survey workflows by performing background data work and digitizing linework in Carlson Survey CAD software.
- Digitized over 100 sewer plans in QGIS to build a new interactive map from scratch, making critical infrastructure data accessible to residents.

Projects

Frapp | NestJS, Next.js, Supabase, React Native (Expo), Stripe

- Architected a multi-tenant SaaS backend with JWT-scoped tenant isolation across three composed guards and row-level security on 50 database tables, validated by a 12-case cross-tenant isolation test suite.
- Built a 20-screen React Native mobile app wired to a 180+ endpoint REST API, reaching 90% test coverage.
- Designed an adversarial AI evaluation harness (43 tests) enforcing citation grounding and prompt-injection defenses ahead of any model implementation.
- Maintained 4,700+ passing tests across a ~300,000-line codebase built solo over 6.5 months.

Tau Nu Fiji Operations Platform | Next.js 14, TypeScript, Appwrite, Discord API, AWS Cloud

- Built a unified platform for a 23-member organization that digitizes facility management and secures academic archives using S3 pre-signed URLs, replacing manual trackers with a centralized architecture.
- Developed a cron-based scheduler that assigns weekly tasks and escalates overdue items from private DMs to public channels, enforcing accountability without manual follow-up.
- Engineered an RBAC system synced with live Discord roles, letting admins manage assignments and penalties directly via native Slash Commands.

Interactive GIS Map | East-SouthEast, LLC Internship

- Delivered a single, dynamic codebase serving 9 towns with JavaScript and Mapbox, replacing a legacy system where every town ran its own one-off code.
- Built a dynamic legend, multi-page PDF report generator, and address search/bookmarking tools used to produce client-facing reports.
- Built a high-performance, cost-effective tile loader for USGS maps, bypassing expensive third-party APIs.

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, SQL

Cloud & Infrastructure: AWS (S3), Supabase, Stripe, Appwrite

Frameworks: Next.js, NestJS, React, React Native (Expo), Node.js

Tools: Git, GitHub Actions (CI/CD), Mapbox GL JS, QGIS

Additional Experience

Camp Counselor & Challenge Course Instructor | Camps Newfound-Owatonna

June – Aug. 2026

- Held 24/7 duty of care for up to 10 campers across two overnight sessions spanning 7 weeks, running a full daily schedule with no adult backup after lights-out.
- Served as one of two certified instructors on the camp's high-ropes challenge course, coaching campers through fear-based challenges and de-escalating peer conflict throughout the summer.

Leadership / Extracurricular

Phi Gamma Delta | Tau Nu Chapter

Chapter President

Jan. 2025 – Present

Jan. 2026 – Present

- Direct an executive cabinet of 5 officers and 10+ committee chairs for a 23-member chapter, managing risk mitigation and relations with university administration and International Headquarters.

Stewardship Chair

Aug. 2026 – Present

- Manage a \$12,000 semester operating budget and weekly procurement logistics for the chapter's residential facility.

VP of Chapter Scholarship

Jan. – May 2026

- Built data-driven academic support systems that kept chapter GPA consistently above the university average.

Eagle Scout | Boy Scouts of America

Nov. 2023

- Led a team to build and install three community benches, managing the project from initial design and fundraising to final installation.